



Review

Technical advancement in valorization of electronic waste and its contribution to establishing economic value-chain

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Highlights

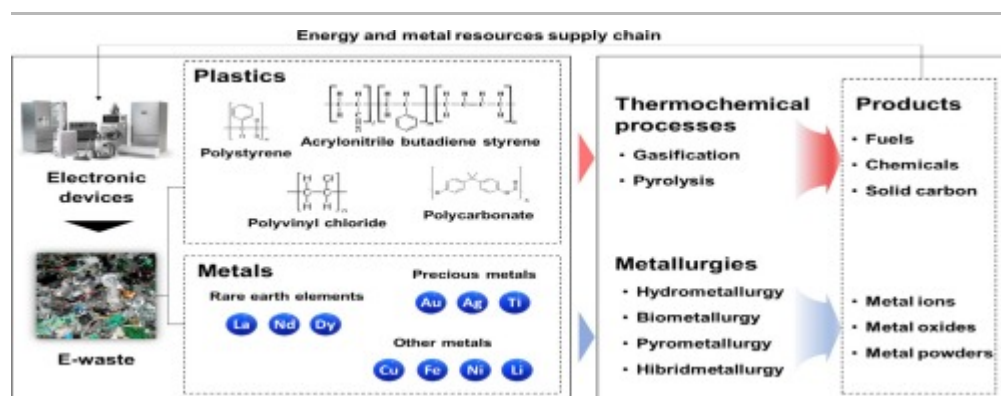
- E-waste generation has been rapidly increasing to threaten human health.
- Current status of global e-waste pollution was scrutinized.
- Reclamation processes for both metallic and non-metallic components were complied.
- Economic value-chain for e-waste can be established through reclamation processes.

Abstract

The rapid expansion of the electronics industry has entailed unavoidable generation of electronic waste (e-waste) in a massive quantity. Although e-waste contains varying

degrees of precious metals and rare earth elements, a substantial portion of e-waste is ended up in landfills and incineration facilities. These management practices preemptively eliminate the possibility of recovering those elements from e-waste and often lead to environmental contamination. Nonetheless, there has been a consistent increase in the global demand for those critical metals, resulting in a steady rise in their prices. To address this issue, researchers have put multifaceted efforts to reclaim e-waste as a renewable resource. Several metallurgical techniques have proven to be effective in refining precious and rare earth metals from e-waste. Besides, thermochemical process has been developed to convert plastic materials in e-waste into fuel gases and other value-added chemicals. These approaches contribute to establishing economic value-chain of e-waste. To this end, this review critically discusses up-to-date technical advancements in e-waste reclamation processes with an emphasis on their potential role in realizing a circular economy. Lastly, the studies on life-cycle assessment of e-waste are reviewed to assess the environmental and industrial consequences of current reclamation processes.

Graphical abstract



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Introduction

The rapidly rising global demand for electronic devices, including electric vehicles, mobile phones, and laptops, has led to a significant increase in the generation of electrical and electronic waste (e-waste) [1], [2]. E-waste is the fastest-growing waste category, with an annual generation reaching 40–50 million tons [3]. The massive generation of e-waste has put significant environmental burdens for its high content of potentially hazardous materials, such as heavy metals and organic chemicals [4], [5]. Hence, appropriate management of e-waste is essentially needed to alleviate its

hazardous environmental impacts. However, a considerable amount of e-waste is still managed by traditional methods such as landfilling and incineration, which are mainly focused on waste volume reduction. These practices result in adverse consequences, including emission of harmful gases into the atmosphere, generation of contaminated wastewater, and soil contamination [6], [7], [8]. Furthermore, implementation of additional pollution control units in existing treatment facilities increases waste management costs, which would undermine the feasibility and accessibility of these approaches for real-world applications [9], [10]. Therefore, there is an urgent need to develop cost-effective and environment-friendly e-waste management strategies to establish a sustainable development of electronics industry.

E-waste contains valuable inorganic components, including rare earth elements (REEs; La, Nd, and Dy), precious metals (Pt, Au, and Ag), and other valuable metals (Li, Ni, and Ti) [11]. However, only 17.4% of these metals are currently being recycled, resulting in an annual wastage of approximately USD 57 billion worth of metals through landfilling, incineration, and other improper processes [12], [13]. To this end, increasing attention has been paid to establishing circular economy platforms for e-waste through technical approaches [12], [14]. In particular, various metallurgies, such as hydrometallurgy, biometallurgy, pyrometallurgy, and hybrid metallurgy, have been applied to develop more efficient metal recovery processes for e-waste [15]. In addition, some researchers have explored thermochemical processes, including gasification and pyrolysis, to convert non-metallic components in e-waste, particularly plastics, into valuable chemicals and other functional materials [16]. Considering that metals and carbon-based chemicals are fundamental resources in existing industries, the sustainable production of these resources from e-waste can be beneficial to entire industries as it can directly reduce supply costs [15].

Owing to the increasing interests in e-waste management strategies, several review articles have compiled previous researches on e-waste management technologies [17]. However, most of these reviews have mainly focused on the metal-recovery processes and that compilation studies on reclamation technologies for non-metallic components in e-waste is relatively scarce. Therefore, this review was prepared to provide a comprehensive overview of up-to-date e-waste management strategies, encompassing the current state of e-waste pollution and the reclamation processes for both metallic and non-metallic components. In addition, research on life cycle assessment (LCA) from different scenarios dependent on e-waste management strategies was reviewed to evaluate their economic and environmental impacts.

Section snippets

Definition and classification of e-waste

The United States Environmental Protection Agency (USEPA) defines e-waste as “any electronic device that has reached the end of its useful life and is no longer wanted for its original purpose” [18]. E-waste is also known as waste electrical and electronic equipment or electronic scrap [19]. To minimize the confusion stemming from the broad meaning of e-waste, many countries have created their own classification systems [20]. However, there is still no global consensus on how to classify ...

Potential values of e-waste

Electronic devices contain various plastics and metals, and their composition varies with the device type, age, and manufacturer [17], [57]. The significant increase in plastic waste has become problematic, largely because it is often improperly managed through landfilling and incineration, leading to severe environmental pollution. Nowadays, many researchers are exploring chemical recycling methods to convert plastic waste into plastic precursors or valuable chemical resources. For instance, ...

Pretreatment step for e-waste valorization

E-waste is a mixture of metallic and non-metallic compounds that requires specific target-oriented recycling processes [88]. Hence, it is necessary to employ appropriate pretreatment steps such as crushing and separation to enhance the recovery rate and minimize energy consumption. An overview of a typical pretreatment process is shown in Fig. 4. The initial stages of e-waste valorization pretreatment involve sorting and dismantling, in which the waste is disassembled into electronics, ...

LCA studies

To evaluate the environmental impacts associated with e-waste, several LCA studies have been conducted that encompass the entire life of e-waste, from production to end-of-life [171], [172], [173]. These studies evaluated and compared the environmental and

economic aspects of several e-waste management scenarios, including direct disposal, end-of-life disposal (landfill and incineration), and value-recovery. Hong et al. evaluated the environmental impact of e-waste disposal processes using ...

Conclusions and future research requirements

The global generation of e-waste has been consistently increasing owing to the expansion of the electronics industry and rapid shifts in market trends of electronic devices. Improper management practices, including landfilling and incineration, contribute to environmental contamination by releasing toxic chemicals and heavy metals into the environment. To address this problem, researchers have strived to investigate metallurgical and thermochemical processes to establish a comprehensive ...

CRedit authorship contribution statement

Gihoon Kwon: Writing – original draft, Conceptualization. **Kwangsuk Yoon:** Methodology, Investigation. **Eilhann Kwon:** Data curation. **Juyeong Park:** Visualization, Validation. **Heuiyun Lee:** Resources, Formal analysis. **Hocheol Song:** Writing – review & editing, Supervision. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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Citation Excerpt :

...This is while recent e-waste monitor report indicates that in the absence of sound e-waste management regulations, a large proportion of e-waste is being handled by informal recyclers or disposed of in landfills (Baldé et al., 2024). Consequently, women and children as part of the informal recycling system in low-income countries as well as soil, water and air near e-waste dumping sites are being contaminated with toxic materials from e-waste (Kwon et al., 2024). Every year, 58,000 kg of mercury and 45 million kg of plastics with BFRs are released into nature from improperly managed e-waste (Baldé et al., 2024)....

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